What is needed for stable synonymy?

Aja Altenhof[*1,2] and Gareth Roberts[2]

[*]Corresponding Author: altenhof@wisc.edu
[1]Department of Linguisticss, University of Wisconsin, Madison, USA
[2]Department of Linguistics, University of Pennsylvania, Philadelphia, USA

An increasingly established view in Language Evolution is that linguistic structure emerges through a process of cultural evolution, where communication provides a pressure for expressivity and iterated learning over generations provides a pressure for compression and reduction in unpredictable variation (Kirby et al., 2015; Smith & Wonnacott, 2010).

One type of structure that has received rather little attention in this literature concerns synonymy. *Absolute* synonymy is known to be rare (Cruse, 1986); synonyms usually exist in structured relationships that carve up semantic space in overlapping but distinct ways (e.g., *immigrant* and *expat*). This pattern could be due to biased sampling from a word's distribution or result from a cognitive bias (e.g., the mutual-exclusivity bias; Lewis et al., 2020). Altenhof and Roberts (2023) investigated this experimentally; participants learned new (synonymous) slang words—which appeared in different contextual distributions—and inserted them into novel sentences. Participants treated the words as not fully synonymous even when training distributions implied they were, suggesting a bias against absolute synonymy. We might expect this pattern to stabilize through iterated learning (Kirby et al., 2014, 2015). Here we present a series of experiments designed to test this. Our results suggest that the emergence of stable, structured synonymy may require relatively robust expressivity pressures.

Method. Following Altenhof and Roberts (2023), participants learned two novel synonymous verbs, *lorgen* and *kiffor*, whose meaning participants had to guess. In the *Exposure* phase, these words were embedded in English sentences with positive, negative, or neutral valence. In the *Generalization* phase, participants inserted them into new sentences. Participants were organized into five-generation iterated learning chains. In the first generation both verbs occurred 50% of the time in positive and negative sentence frames. Differentiation was calculated using *Gini scores* (Abounoori & McCloughan, 2003; Roberts & Galantucci, 2016). While single-generation results replicated Altenhof and Roberts (2023), chains remained very unstable (Figure 1). The same results replicated in a series of follow-up experiments investigating factors that might encourage

or discourage stability, including providing an explicit meaning for the two verbs ("do graffiti"), doubling the number of sentences in both phases, and employing an incremental reading paradigm to increase attention.

Results and Discussion. Data are available at https://osf.io/mf9xp/. While participants never treated the words as absolute synonyms, a *stable* pattern of structured synonymy did not emerge in any experiment (Figure 1); generation main effect: $t(28.4) = 0.91, p = .37$; generation $\times$ version interactions: all $ps > 0.15$)[1]

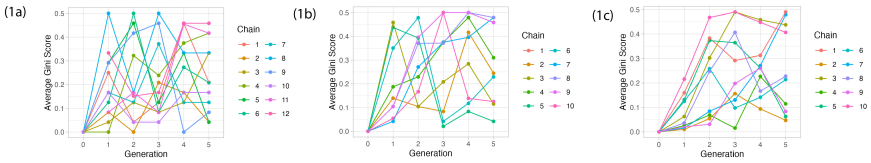

Figure 1. Participants Gini scores by generation (1) for the graffiti (a), doubled sentences (b), and incremental reading (c) versions of the tasks.

These results seem to conflict with prior literature, where iterated learning led to the reduction of unpredictable variation, such that we might expect it to lead to structured synonymy (Smith & Wonnacott, 2010). But this does not take into account the role of expressivity pressures from communication. Our experiment used a proxy for communication, which was apparently sufficient to motivate participants to employ and distinguish both verbs (only $< 4\%$ of all participants dropped one of the verbs). However, inserting words into a blank is not a very strong communicative pressure. We argue that real interaction may be required to motivate stable, structured synonymy and will present results from ongoing experiments testing precisely this.

In sum, while iterated learning and a weak communicative pressure were sufficient to discourage absolute synonymy, they were insufficient for the emergence of stable structure. Ongoing work with more robust participant interaction aims to clarify the role of the expressivity pressure in this process.

References

Abounoori, E., & McCloughan, P. (2003). A simple way to calculate the gini coefficient for grouped as well as ungrouped data. *Applied Economics Letters*, *10*(8), 505–509.

[1]We fit a linear mixed-effects model with random intercepts and random slopes for generation by chain. The model included generation, version (graffiti, doubled sentences, or incremental reading), and their interaction as fixed effects. Generations within chains are temporally dependent; however, autocorrelation would increase standard errors and reduce statistical power, making our findings more conservative.

Altenhof, A., & Roberts, G. (2023). Quality, not quantity, impacts the differentiation of near-synonyms. *Language and Cognition*, *15*(4), 854–883.

Cruse, D. A. (1986). *Lexical semantics.* Cambridge University Press.

Kirby, S., Griffiths, T., & Smith, K. (2014). Iterated learning and the evolution of language. *Current Opinion in Neurobiology*, *28*, 108–114.

Kirby, S., Tamariz, M., Cornish, H., & Smith, K. (2015). Compression and communication in the cultural evolution of linguistic structure. *Cognition*, *141*, 87–102.

Lewis, M., Cristiano, V., Lake, B. M., Kwan, T., & Frank, M. C. (2020). The role of developmental change and linguistic experience in the mutual exclusivity effect. *Cognition*, *198*, 104191.

Roberts, G., & Galantucci, B. (2016). Investigating meaning in experimental semiotics. *Psychology of Language and Communication*(2), 130–153.

Smith, K., & Wonnacott, E. (2010). Eliminating unpredictable variation through iterated learning. *Cognition*, *116*(3), 444–449.